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Review article

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Harnessing artificial intelligence to advance CRISPR-based genome editing technologies

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Abstract

CRISPR-based genome editing technologies, including nuclease-based editing, base editing and prime editing, have revolutionized biological research and modern medicine by enabling precise, programmable modification of the genome and offering new therapeutic strategies for a wide range of genetic diseases. Artificial intelligence (AI), including machine learning and deep learning models, is now further advancing the field by accelerating the optimization of gene editors for diverse targets, guiding the engineering of existing tools and supporting the discovery of novel genome-editing enzymes. In this Review, we summarize key AI methodologies underlying these advances and discuss their recent noteworthy applications to genome editing technologies. We also discuss emerging opportunities, such as AI-powered virtual cell models, which can guide genome editing through target selection or prediction of functional outcomes. Finally, we identify key directions where the integration of AI methods is poised to have a substantial impact going forward.

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Introduction

Genome editing has emerged as one of the most transformative advances in modern biotechnology. The ability to precisely and programmably modify DNA sequences in living organisms has revolutionized basic research and enabled new applications in agriculture, medicine and synthetic biology[1–3]. Since the discovery of CRISPR–Cas9 as an RNA-guided nuclease[4], the field has rapidly expanded, unlocking therapeutic potential for a wide range of previously untreatable genetic disorders[5]. Recent innovations such as base editing and prime editing now allow for precise nucleotide conversions and targeted insertions or deletions (indels) without inducing DNA double-strand breaks (DSBs), offering a theoretical path to correct more than 90% of known pathogenic mutations[6].

At their core, genome editing systems are modular assemblies composed of a programmable nuclease, a targeting mechanism, which is typically a guide RNA (gRNA) or engineered DNA-binding protein, and often additional elements that modulate repair outcomes[7] (Fig. 1). The prototypical CRISPR–Cas9 system[4] exemplifies this architecture: a Cas9 nuclease complexed with a gRNA targets DNA sequences adjacent to a protospacer adjacent motif (PAM) to cleave specific genomic loci. More advanced editors, such as base editors (BEs)[8] and prime editors (PEs)[6], augment this framework by fusing Cas proteins with enzymatic effectors (for example, deaminases or reverse transcriptases (RTs)) and by incorporating modified gRNAs to direct precise sequence changes. Historically, genome editing tools have been developed through a combination of natural discovery and protein engineering. Prior to CRISPR, targeted endonucleases such as zinc-finger nucleases (ZFNs)[9] and transcription activator-like effector nucleases (TALENs)[10] required laborious re-engineering for each new genomic site. CRISPR–Cas9 revolutionized this process by enabling flexible, RNA-guided targeting through simple alteration of the gRNA sequence. Since then, systematic mining of microbial genomes has yielded a broad spectrum of Cas nucleases and effectors, including Cas12 for DNA targeting, Cas13 for RNA targeting, and transposase-based and recombinase-based systems that enable editing without inducing DSBs[11]. Concurrently, the optimization of editing activity and specificity has relied heavily on structure-guided mutagenesis and directed evolution[12]. These approaches have produced Cas9 variants with improved fidelity and expanded PAM compatibility, such as those engineered using phage-assisted continuous evolution[13].

Despite these advances, the efficient design, optimization and application of gene editors still faces challenges. Editing outcomes are highly context-dependent, influenced by the gRNA design, target sequence features, chromatin accessibility and cellular environment[7]. Navigating this complex design space poses substantial hurdles for conventional experimentation or expert intuition. Furthermore, the optimization of genome editing systems, which has traditionally relied on protein engineering, remains constrained by limited structural data and the resource-intensive nature of high-throughput mutagenesis and screening[12]. Traditional bioinformatic pipelines for enzyme discovery, typically reliant on sequence homology, struggle to identify novel effectors in vast, unannotated metagenomic datasets, particularly when candidate proteins lack similarity to known enzymes[14,15]. Consequently, both the optimization of existing genome editors and the discovery of new ones remain slow, empirical and incomplete[16].

Artificial intelligence (AI) offers a powerful framework to address these challenges by enabling predictive modelling, functional inference and data-driven design throughout the genome editing pipeline (Box 1). Machine learning models trained on large-scale datasets can predict gRNA activity, target accessibility, editing outcomes and off-target risks, thereby improving the efficiency and precision of gRNA design. In parallel, AI is accelerating the engineering of editor proteins by modelling sequence–function relationships and proposing beneficial mutations that enhance activity, broaden PAM compatibility or reduce off-target effects. Deep learning-based protein structure prediction tools, such as AlphaFold[17–19] and RoseTTAFold[20], have democratized access to high-resolution structures, including for genome editing enzymes such as Cas nucleases and deaminases. This structural insight facilitates more rational protein design and supports iterative variant optimization. Beyond predictive modelling, generative AI approaches are being deployed for de novo design of functional biomolecules, including synthetic nucleases, engineered deaminases and structured RNA scaffolds optimized for enhanced stability, specificity or modularity[21].

In this Review, we provide a brief overview of current CRISPR-based genome editing tools and then discuss how AI is advancing the design, optimization and scalability of these technologies. We highlight how AI is being applied to improve key components of genome editing systems, including gRNA selection, editor protein engineering and the identification of novel effectors, across CRISPR–Cas nucleases, BEs and PEs. We also discuss opportunities and challenges in applying AI to accelerate therapeutic translation and expand the applicability of these tools.

CRISPR-based genome editing tools

The bulk of modern genome editing tools fall under the umbrella of RNA-guided CRISPR–Cas endonucleases, initially discovered as an adaptive bacterial defence system against invading phage DNA[11]. The RNA-directed targeting mechanism was readily recognized and repurposed for RNA-programmable cleavage of essentially any DNA sequence[4], laying the foundation for a genome editing system with unmatched versatility.

Nuclease editing

Initial applications of CRISPR–Cas systems for genome editing used Cas nucleases to introduce DSBs at genomic regions specified by the spacer sequence of an associated single guide RNA (sgRNA)[11] (Fig. 1a). The approximately 20-nucleotide spacer sequence positions the Cas nuclease at a DNA target through complementary base pairing, provided a suitable PAM sequence, which is required for Cas binding, is directly downstream (3′) of the target site. Once bound, the nuclease domain of the Cas enzyme will cleave the DNA, generating a DSB. A DSB is typically resolved by non-homologous end joining (NHEJ) or microhomology-mediated end joining (MMEJ), leading to random indels at the cut site[22], which can be harnessed for gene disruption (knockout). Alternatively, if a donor DNA template with regions of homology to the cut site is provided, DNA repair can proceed through homology-directed repair, allowing targeted insertion of the desired sequence at the target site, although indels still occur.

The Cas9 family of enzymes is most often used for genome editing applications, particularly SpCas9 from *Streptococcus pyogenes*, owing to its high editing efficiency and more common PAM sequence preferences[11,23]. The PAM sequence recognized by SpCas9 is NGG (where N denotes any nucleotide and GG is two guanines), although PAM requirements vary between Cas orthologues. Cas12 enzymes, which generate staggered DSBs at T-rich PAMs, have also been repurposed for genome editing[24,25]. Although their native editing efficiency is lower than that of SpCas9, engineered variants of Cas12 have demonstrated comparable performance[26]. Enzymes in the Cas13 family can be used to target RNA instead of DNA[7,27], enabling gene knockdown at

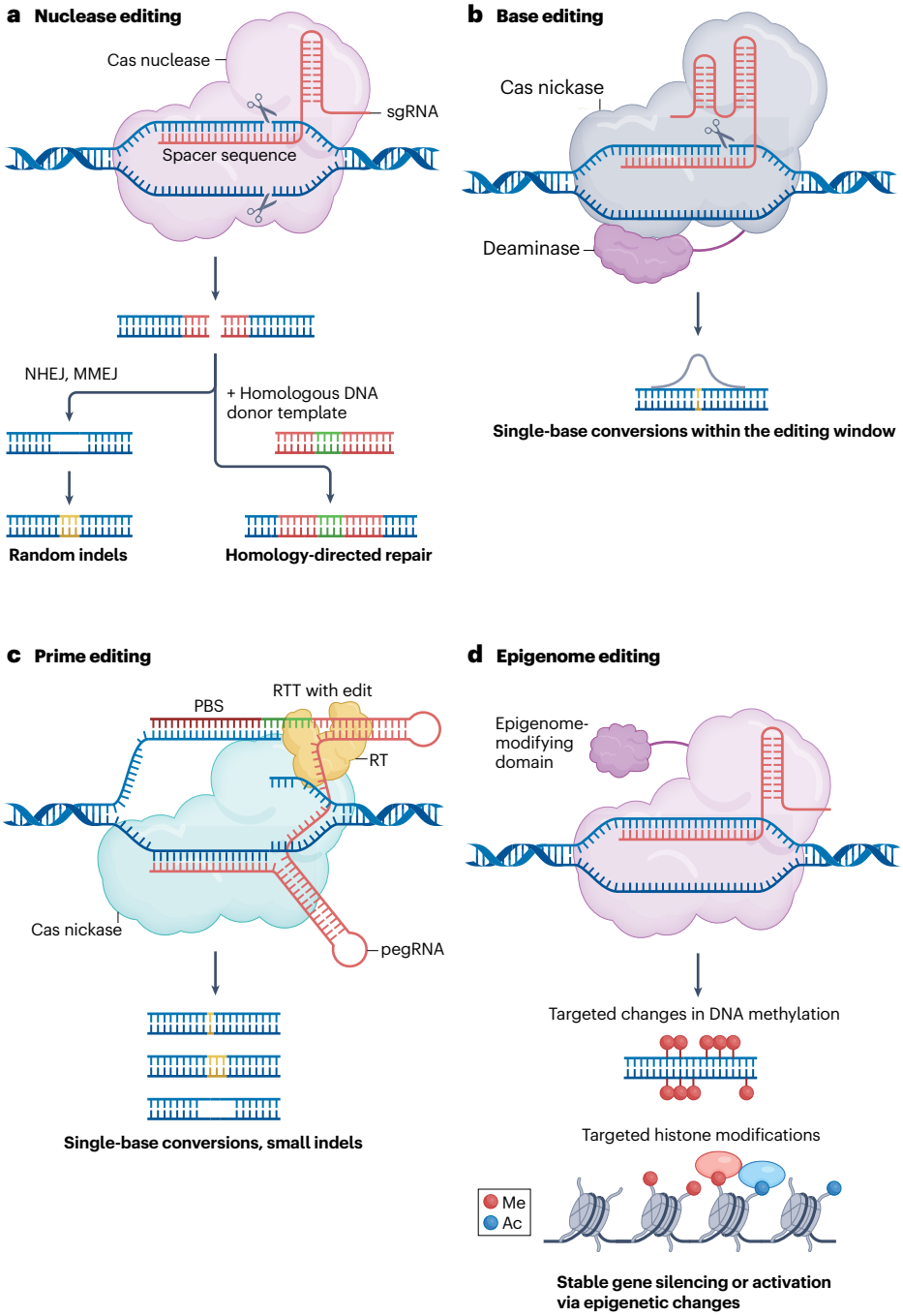


Fig. 1 | The core suite of modern CRISPR-based genome editing tools. a, Nucleases initiate genome editing through generating double-strand breaks (DSBs) in the genome at sites specified by the spacer sequence of the single guide RNA (sgRNA). DSBs will result in the generation of random insertions or deletions (indels), or additional incorporation of a DNA template with homology-directed repair. **b**, Base editors (BEs) consist of a nuclease-deficient Cas enzyme with an attached base-modifying enzyme (typically a deaminase). Upon binding to the target site, the deaminase can act on accessible single-stranded DNA bases (within its editing window) enabling precise base conversions. **c**, A prime editor (PE) consists of a Cas nickase with an attached reverse transcriptase (RT) domain. Upon nicking, the nicked DNA strand can anneal to the complementary primer binding site (PBS) of the prime editing guide RNA (pegRNA), serving as a primer for the RT to incorporate the edit specified by the pegRNA RT template (RTT). **d**, Cas enzymes can be used for epigenetic editing through the attachment of enzyme domains that influence DNA methylation or histone modifications, facilitating stable changes in gene expression without modifying the genome. MMEJ, microhomology-mediated end joining; NHEJ, non-homologous end joining.

the RNA level, an advantageous feature to avoid inducing permanent genomic changes. Recently discovered ancestral Cas-related proteins, such as the transposon-encoded TnpB and IscB nucleases, which use an associated ωRNA for target recognition, are emerging as programmable genome editing tools[28–30]. These enzymes benefit from having a more compact size (approximately 400 versus 1,368 amino acids for SpCas9) but will require future engineering to increase editing efficiencies. Notable progress is being made, with engineered TnpB[29] and IscB[31] enzymes achieving respective editing efficiencies of approximately 0.3× and 0.8× that of SpCas9 at equivalent targets. IscB also shows strong potential for RNA editing, as disruption of its DNA-binding domain resulted in preferential RNA targeting at levels outperforming Cas13, with an additional benefit of reduced cytotoxicity[32].

The occurrence of a DSB limits the utility of traditional CRISPR–Cas nucleases for broader applications, such as the precise correction of disease-causing mutations. DSBs can cause several undesired

Box 1 | Artificial intelligence techniques for genome editing applications

Machine learning

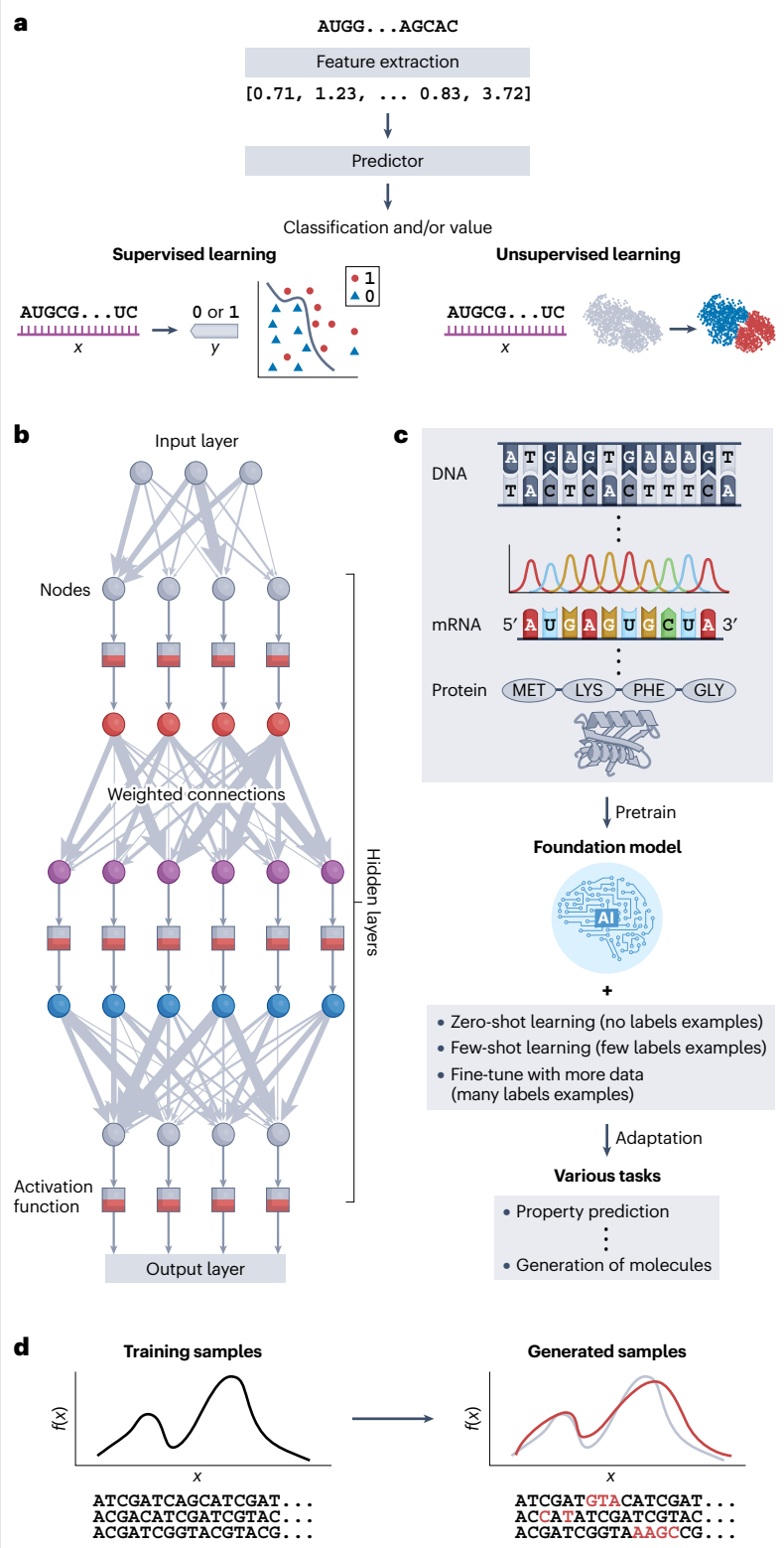
Machine learning enables computational systems to identify patterns and make predictions based on large datasets without explicit programming[171] (see the figure, part **a**). Machine learning models are typically composed of two components: feature extractors, which identify informative representations of the input data; and predictors, which map these features to outputs. Unlike traditional rule-based programming, machine learning systems iteratively improve their performance by adjusting internal parameters through exposure to data. The training process begins by fitting the model to a labelled training dataset, allowing it to learn statistical relationships between inputs and outputs through optimization of a loss function. A separate validation dataset is then used to fine-tune hyperparameters and monitor overfitting, ensuring the model generalizes well beyond the training set. Finally, an independent test dataset is used to evaluate model performance under realistic conditions. This safeguards against data leakage, a situation where information from the training set is inadvertently used during evaluation, ensuring the model's predictive ability truly reflects its capacity to generalize to unseen data.

Deep learning

Deep learning, a subfield of machine learning, has emerged as a powerful tool for extracting complex, hierarchical patterns from biological data (see the figure, part **b**)[172]. Unlike traditional machine learning methods, which often rely on handcrafted features, deep learning models learn representations directly from raw data, enabling the analysis of large and complex biological datasets with minimal human input. At the core of deep learning are artificial neural networks composed of interconnected layers of computational nodes ('neurons') that transform input data via weighted connections and non-linear activation functions. When multiple layers are stacked into a deep neural network, the model gains the capacity to learn increasingly abstract representations from raw data, enabling it to capture subtle, hierarchical or long-range biological patterns. Two commonly used neural network architectures are convolutional neural networks (CNNs) and recurrent neural networks (RNNs)[173]. CNNs use sliding convolutional filters to identify spatial patterns in data, such as nucleotide motifs associated with increased editing efficiency, whereas RNNs can recall information in previous processing steps, making them well suited to sequential data.

Foundation models

Foundation models are large-scale, pretrained deep learning models designed to be adaptable across diverse downstream tasks[174] (see the figure, part **c**). Trained using self-supervised learning on vast, unlabelled datasets, these models learn universal representations



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of biological sequences or structures that can be fine-tuned for specific applications with minimal additional data. A key advantage of foundation models is their support for zero-shot and few-shot learning. In zero-shot learning, the model generalizes to a new task directly from its pretrained knowledge, without requiring additional fine-tuning. In few-shot learning, the model can be rapidly adapted to novel tasks using only a small number of labelled examples. These capabilities are particularly valuable in biology, where many problems are underexplored and annotated datasets are limited.

Generative artificial intelligence

Generative artificial intelligence (AI) refers to a class of models that learn the underlying distribution of training data and can generate novel data samples that resemble the training distribution without direct reference to specific input examples[175] (see the figure, part **d**). Generative AI enables the creation of entirely new biological sequences or structures with desired features, making it a powerful tool for de novo molecular design. Language models have become the dominant architecture for sequence generation due to their scalability, ability to incorporate long-range dependencies, and compatibility with fine-tuning and conditioning tasks[21]. Many state-of-the-art generative models in biology also fall under the category of foundation models, large-scale models pretrained on diverse datasets to enable generalization across downstream tasks. For example, ESM-3, which has been trained on millions of protein sequences and structures with functional annotations, functions both as a foundation model, owing to its large-scale pretraining, and as a generative model, given its capacity to design entirely novel proteins[176]. As the quality and quantity of biological training data continue to improve, and as generative model architectures evolve, generative AI is poised to play a central role in the next generation of genome engineering tools.

consequences, including induction of detrimental DNA damage response pathways[33], large-scale chromosomal rearrangements[34] and chromothripsis[35], motivating the development of editing systems that can act in a DSB-independent manner.

Base editing

The first major development for genome editing without DSBs was base editing, which harnesses a nuclease-deficient Cas protein (most often a nickase; that is, only one of the two Cas nuclease domains is mutated) with an attached deaminase enzyme[7] (Fig. 1b). Upon Cas binding to the target site specified by the sgRNA, the deaminase enzyme can act on the exposed bases of single-stranded DNA within a typically 5 bp editing window, enabling targeted base conversions. Most BEs are cytosine BEs (CBEs)[36] or adenine BEs (ABEs)[37], which catalyse C-to-T and A-to-G base changes, respectively. Incorporating glycosylase domains into the BE architecture can further expand this editing scope to include transversion edits such as C-to-G, A-to-C/T and G-to-C/T, although with varying product purity[38]. Attaching deaminase enzymes to RNA-targeting Cas proteins such as Cas13 also enables targeted editing of the cellular transcriptome[7]. BEs benefit from high on-target efficiency and are suitable for many therapeutically relevant targets, already showing striking success in preclinical and clinical settings for several indications[39,40]. However, BEs have some important limitations that prevent their use as a universal genome editing tool, as they currently cannot perform all base conversions and will edit multiple copies of the same base if present within the editing window (known as bystander editing), potentially leading to undesirable editing outcomes[7]. Additionally, the high activity of deaminase enzymes can pose a risk of off-target editing within the genome and transcriptome[7,41].

Prime editing

The development of prime editing helped overcome some of the limitations of base editing. A PE consists of an RT fused to a Cas9 nickase (nCas9), which is directed to target sites by an associated prime editing guide RNA (pegRNA)[42] (Fig. 1c). The pegRNA differs from a typical sgRNA by including two additional elements at its 3′ end, a primer binding site (PBS) and an RT template (RTT). The PBS is complementary to the nCas9-nicked DNA strand and acts as a primer for the RT to write in an edit specified by the RTT. A secondary nicking gRNA is also often included to nick the opposing non-edited strand and bias DNA repair towards incorporation of the desired edit[42]. PEs can catalyse all base substitutions, as well as small indels, enabling precise correction of more than 90% of known pathogenic disease variants[6]. Larger-scale genomic changes can also be performed by using the PE to write in recombinase recognition sequences, at which large DNA fragments can be inserted when provided alongside the appropriate recombinase enzyme[43–45]. Although prime editing benefits from exceptional versatility and typically negligible levels of off-target editing[6], its editing efficiencies are not yet on par with base editing, and it also requires an extensive optimization process for effective editing at different targets[46]. Recent efforts aim to address this limitation by engineering more effective PEs[47–49] and pegRNAs[47,50–52].

Epigenome editing

CRISPR-based systems have also been harnessed to induce targeted changes in the epigenome by fusing epigenetic-modifying domains to a Cas enzyme lacking any nuclease activity (‘dead’ Cas9 (dCas9); that is, both nuclease domains of the Cas protein are mutated)[7]. The principle of sgRNA-guided targeting can then be used to induce epigenetic modifications at specific genomic sites, allowing durable inhibition or activation of gene expression without changes to the genome (Fig. 1d). Initial efforts in this domain focused on modulating transcriptional activity with the development of CRISPR activation (CRISPRa) and CRISPR inhibition (CRISPRi)[53]. CRISPRa relies on the fusion or recruitment of transcriptional activation domains[54,55], such as HSF1, VP64 or Rta, to dCas9, whereas CRISPRi consists of dCas9 fused to a KRAB repressor[54]. Although effective, CRISPRa and CRISPRi only alter transcriptional activity transiently, with a return to baseline transcription levels once absent from the cell, motivating the need for systems capable of inducing more permanent changes. The CRISPRoff system addressed this challenge by attaching KRAB and DNA methyltransferase domains to dCas9 to induce long-term changes[56]. These edits could be reversed by the CRISPRon system, which involves fusion of a dCas9 to a DNA demethylase, along with recruitment of additional transcriptional activator proteins[56]. Epigenetic gene activation has also been achieved using dCas9 fusions with enzymes such as histone acetylases[57] and kinases[58].

Artificial intelligence for gRNA optimization

Selecting the appropriate gRNA for a given target is a crucial first step to achieve optimal genome editing. Editing outcomes are highly dependent on the gRNA spacer sequence, which can vary considerably across its approximately 20-nucleotide length. Identifying the features and patterns — such as nucleotide composition, positional nucleotide preferences and secondary structures[59] — associated with increased editing efficiency within this sequence space is especially well suited to AI methods, which has fuelled the development of sgRNA prediction models as one of the first AI applications for genome editing. Numerous different machine learning and deep learning models have been trained by screening large numbers of sgRNA spacers in vitro or using available datasets of editing outcomes. In the following sections, we highlight notable recent examples and refer readers to specific reviews on this application for additional details[59–61].

Prediction models for nuclease sgRNA

The development of techniques for high-throughput sgRNA screening enabled the efficient generation of large-scale datasets necessary for effective training of sgRNA activity prediction models. Two key experimental methods to measure sgRNA activity at scale are the library-on-library approach[62] and the pairwise library approach[63]. In the library-on-library approach, cells are transduced with a library of target sequences using lentiviruses, followed by transient transfection with a library of sgRNA-expressing plasmids, to determine which sgRNAs effectively cut each target. The pairwise library approach uses lentiviral constructs that encode both the target and corresponding sgRNA, and has been more widely adopted due to its high accuracy and compatibility with large-scale screens.

Such large-scale datasets have been used to train prediction models using simpler machine learning architectures. Data from the library-on-library method were initially used to train the sgRNA Scorer support vector machine model for Cas9 sgRNA prediction based on nucleotide distribution throughout the spacer sequence[62]. Similarly, the CINDEL model was developed from the pairwise library dataset and used logistic regression based on nucleotide position preferences and GC content for Cas12a sgRNAs[63]. Although showing some success in sgRNA prediction, these tools relied on manual feature identification (for example, GC content, melting temperature), which fails to recognize more complex patterns such as position-dependent base interactions within the sgRNA sequence. Deep learning sgRNA prediction models overcome this problem by enabling automatic learning of sgRNA features associated with increased editing efficiency without requiring manual input[64] (Fig. 2a).

DeepCpf1 is a pioneering deep learning model developed to predict sgRNA activity for Cas12a (which was previously known as Cpf1)[65] based on sgRNA activity at its intended target site. This model, which uses convolutional neural networks (CNNs) to automatically learn complex sequence patterns, was trained on data from a screen of more than 15,000 Cas12a sgRNAs in HEK293T and HCT116 cells. DeepCpf1 demonstrated improved generalizability on external test datasets compared with earlier models and provided the first compelling evidence that incorporating cell line-specific epigenetic features, such as chromatin accessibility, can improve the prediction of genome editing activity at endogenous loci. Using a similar strategy, the same group developed DeepSpCas9, a model trained on a screen of more than 12,000 SpCas9 sgRNAs[66]. Similar to DeepCpf1, DeepSpCas9 exhibited strong predictive accuracy and broad generalizability across independent datasets.

Building on these developments, subsequent models have leveraged pre-existing datasets to expand training scale. DeepCRISPR, for example, was trained on approximately 200,000 sgRNAs with known knockout efficiencies[67]. More recently, CRISPRon[68] combined in vitro sgRNA screening with publicly available datasets to develop a deep learning model for SpCas9 activity prediction.

Expanded deep learning approaches are proving beneficial for Cas9 sgRNA activity prediction. CRISPR_HNN supplements a deep learning model with additional architectural features, such as a bidirectional gated recurrent unit, multiscale convolution, multi-head self-attention and one-hot encoding mechanisms, which improved prediction accuracy and highlighted the impact of position-specific base substitutions within the spacer sequence[69]. Another recent deep learning model, PLM-CRISPR, predicts sgRNA activity across different Cas9 variants[70]; by incorporating the amino acid sequence of a given Cas9 protein through the ESM-2 protein language model[71], PLM-CRISPR achieves strong predictive performance and enables generalization to Cas9 variants lacking extensive training datasets. DeepCRISTL[72] improves sgRNA activity prediction by using transfer learning to fine-tune the CRISPRon[68] model on smaller, cell context-specific datasets. Beyond Cas enzymes, TnpB editing efficiency predictor (TEEP) is a deep learning model trained on approximately 10,000 different target sites to predict the nuclease activity of TnpB-guiding ωRNAs[29].

Off-target editing (editing events at sites other than the target sequence) is another key consideration for sgRNA selection. Prediction models can aid in the initial selection of an sgRNA by filtering out candidates with a higher likelihood of off-target editing. Several deep learning models, such as DeepCRISPR[67] and CnnCRISPR[73], have enabled off-target efficiency prediction by screening sgRNAs with mismatched bases or using existing databases of reported off-target edits[59]. More recently, CRISOT additionally incorporated molecular dynamics simulations to predict off-target edits, demonstrating up to 2-fold improved performance over previous methods[74]. A further enhancement in off-target prediction came with the development of crispAI, which provides uncertainty estimates for predicted edits rather than a single-point score of editing likelihood, thereby providing researchers with a more informed overview of potential off-target activity[75]. Using *k*-mer embeddings for DNA sequence representation was also found to increase off-target prediction accuracy in the CRISPR-Embedding model[76]. Language models are increasingly being applied to off-target prediction as well. CCLMoff is a deep learning model trained on deep sequencing data from 13 genome-wide off-target assays and incorporates a pretrained RNA language model to better estimate the likelihood of sgRNA binding at off-target sites, enabling improved predictive performance across diverse test datasets[77]. Deep learning has also been used to predict the on-target and off-target activities of the RNA-targeting Cas13d, for example with DeepCas13 (ref. 78), which was trained on approximately 23,000 screened and previously published Cas13d sgRNAs, and TIGER[79], which demonstrated improved performance over previous models by using a screen of approximately 200,000 sgRNAs for model training.

Finally, DNA repair outcomes downstream of sgRNA activity can also be predicted with machine learning models such as inDelphi[80], Lindel[81], FORECasT[82] and SPROUT[83], which estimate the likelihood of indels at Cas9 cut sites. Building on inDelphi predictions, a study recently demonstrated that designing the ends of the donor DNA to match microhomology sequences predicted at CRISPR cut sites improved its integration via MMEJ[84].

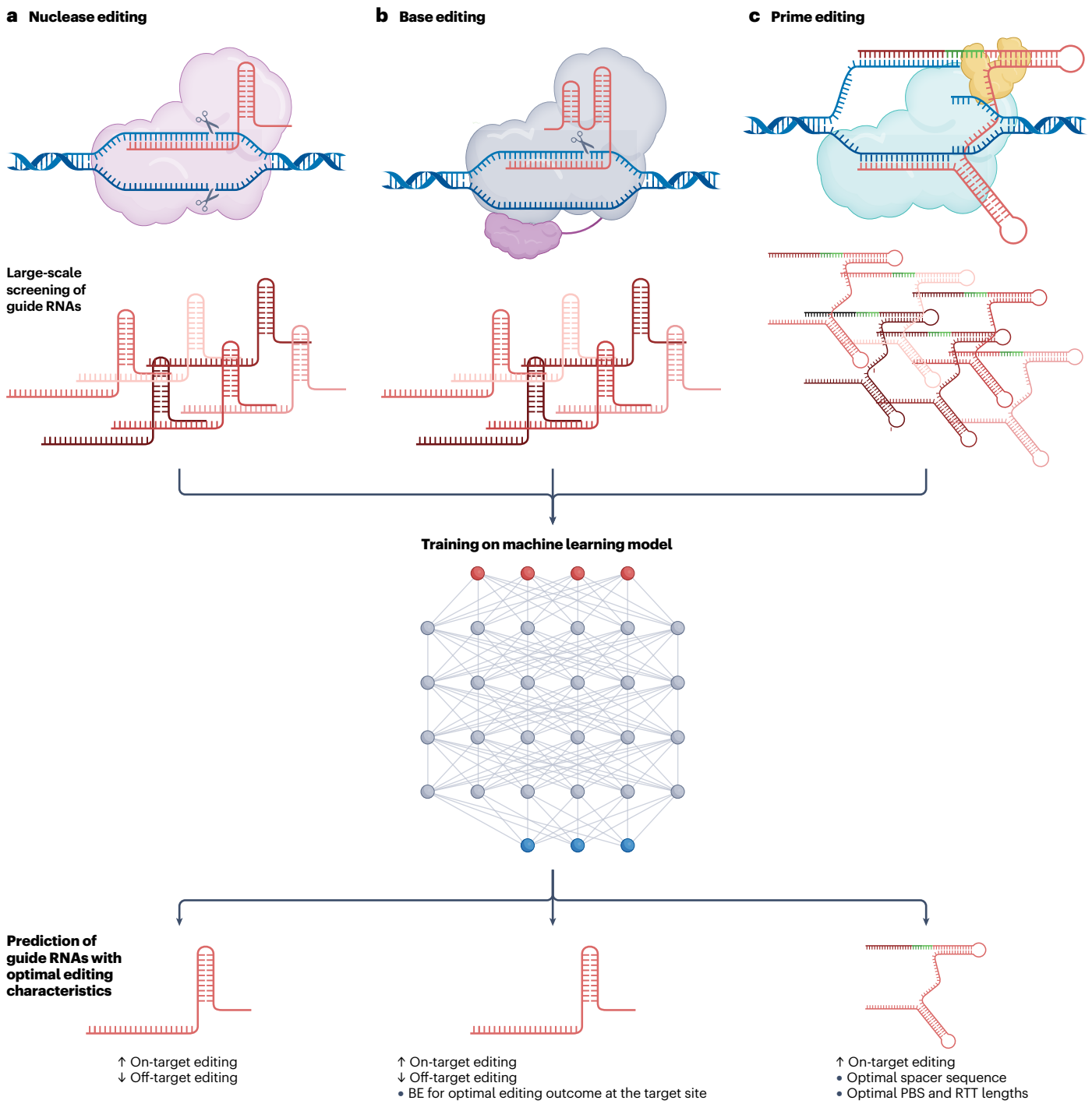

Fig. 2 | Machine learning prediction models for the identification of optimal guide RNAs. Large screens of guide RNAs (gRNAs) for different editing modalities are conducted to generate data indicating editing efficiency for each gRNA sequence. Alternatively, results from existing datasets may be used for model training. Results are provided to a machine learning or deep learning algorithm, which can identify gRNA sequence features associated with desired editing outcomes. At a given target, researchers can use the prediction models to generate a gRNA with a higher likelihood of containing the optimal characteristics for their application. **a**, Nuclease editing models allow identification of single guide RNAs (sgRNAs) with expected high on-target activity and lowered chances off-target editing. **b**, In addition to on-target and off-target editing prediction, models for base editing can incorporate information about the target site to aid in the selection of the optimal base editor (BE) to use. **c**, Prime editing guide RNA (pegRNA) prediction models aid in prime editing optimization by also identifying primer binding site (PBS) and reverse transcriptase (RT) template (RTT) lengths predicted to have increased editing efficiencies.

Prediction models for base editor sgRNA

Although BEs still use an sgRNA to direct them to their target site, additional factors need to be considered for optimal editing, such as the BE editing window, the position of the target base within it and the nucleotides surrounding the target base[85]. As with nuclease editing, several studies have harnessed large-scale sgRNA screening in conjunction with deep learning methods to better understand and predict editing outcomes (Fig. 2b).

An initial success in using deep learning for BE prediction was BE-Hive, which was trained on 38,538 sgRNAs in 3 mammalian cell types using 11 different ABEs and CBEs[86]. BE-Hive was capable of accurately predicting editing efficiencies and specific editing outcomes, including bystander editing and rare CBE transversion edits (C-to-A or C-to-G). The concurrently developed DeepABE/DeepCBE[87] and the later developed BE-DICT[88] deep learning models demonstrated comparable performance in predicting outcomes on test datasets[88], although both are applicable to a smaller set of BEs. In later work, BE-Hive was expanded to predict the activity of C-to-G editors, using a dataset of editing outcomes from 10 different C-to-G BEs at 10,638 targets to develop CGBE-Hive[89].

The utility of BE prediction models was further enhanced with the development of DeepBE, which was trained with seven different BEs (including ABEs, CBEs and C-to-G BEs) and incorporated DeepCas9variants to predict the activity of nine different Cas9 PAM variants[90]. DeepBE enables prediction of both editing efficiency and the optimal Cas9 PAM variant for a particular target, with BE strategies suggested by DeepBE consistently outperforming those based on rational design. Additionally, the deep learning model BEguider[91] expanded the application of prediction models to near-PAMless ABEs and CBEs that use the SpRY Cas9 variant[92].

Incorporating factors outside the target DNA sequence has also been shown to improve base editing outcome predictions. For example, integrating additional target site features such as DNA methylation, histone modifications and transcriptional activity into a deep learning model improved prediction accuracy at endogenous targets[93].

Finally, prediction models to predict BE off-target edits include ABEdeepoff and CBEdeepoff[94]. sgRNAs with various mismatches were screened relative to their target site to obtain editing outcomes for more than 50,000 edits with both an ABE and a CBE, which were then used to train respective deep learning models capable of predicting editing efficiency at a given off-target sequence[94].

Prediction models for pegRNA optimization

The expanded function of the pegRNA adds additional complexity to optimizing editing efficiency, requiring selection of an optimal spacer sequence and PBS and RTT lengths, all of which can vary substantially across different targets[46]. To address this challenge, several groups have trained deep learning models with large libraries of pegRNAs to identify sequence features associated with higher levels of editing (Fig. 2c).

Early efforts focused on predicting single base substitutions, including DeepPE[95], trained for a single edit type (a G-to-C conversion 5 bases from the nick site), and EasyPrime[96], a model trained on more than 46,000 editing outcomes from published datasets for various edits. Improvements were made in later models by expanding the size of training datasets, as seen with PRIDICT[97], which used 92,423 pegRNAs targeting 13,349 pathogenic point mutations, and DeepPrime[98], where 300,000 pegRNAs were screened to train a prediction model applicable for up to 3 bp substitutions and different editing positions.

More recent work has enabled prediction in editing contexts beyond base substitutions. PRIDICT2.0 used a library of pegRNAs encoding more diverse edit types, such as multiple base replacements and small indels[99], and also incorporated epigenetic data such as histone modifications and chromatin accessibility to predict editing efficiency across different epigenetic contexts with ePRIDICT. Additionally, MinsePIE focuses on predicting the efficiency of small insertions of up to 69 nucleotides[100].

AI methods beyond traditional machine learning and deep learning approaches are also being adopted for pegRNA prediction. Optimized Prime Editing Design (OPED) is a nucleotide language model using deep transfer learning that was found to consistently outperform previous prediction models on various datasets[101]. More recently, DTMP-Prime[102] uses a deep transformer-based model to predict the efficiency of pegRNAs and nicking gRNAs, as well as off-target likelihood, incorporating features such as RNA secondary structure and pegRNA–DNA sequence interactions.

sgRNA prediction models for epigenome editing

In contrast to the above genome editing modalities, substantially less attention has been given to developing sgRNA prediction tools for epigenome editing. Compared with DNA editing, epigenetic regulation involves more diverse and context-dependent modifications, which complicates the prediction of functional outcomes. For example, the impact of an epigenetic modification on gene expression at a target site will also depend on the surrounding epigenetic landscape and genomic position[103], factors that can also influence epigenome editing efficiency[104]. The scale of factors involved here provides substantial complexity; nevertheless, AI methods can effectively identify patterns within large quantities of data.

An initial step towards this goal came with the development of EpiCas-DL[105], a deep learning model trained using published datasets for CRISPRa, CRISPRi and CRISPRoff, which integrates epigenetic features such as histone modifications and chromatin accessibility. Across these modalities, EpiCas-DL exhibited greater accuracy than previously developed sgRNA prediction models for Cas9 nuclease activity. The launch-dCas9 model was later developed for CRISPRi using a dataset of more than 1 million sgRNAs targeting more than 100,000 *cis*-regulatory elements[106]. In addition to sgRNA sequence, launch-dCas9 incorporates thermodynamic properties, target site epigenetic marks and gene essentiality to predict the impact of an sgRNA on cell fitness, gene expression and sgRNA abundance. Another approach used histone modification and RNA sequencing data from the ENCODE consortium[107] to develop a deep learning model for gene expression prediction[108]. By modelling the likelihood of dCas9-P300 activity, which increases histone H3 acetylated at K27 (H3K27ac), at a given site, the authors were able to successfully predict changes in gene expression across eight different genes. Beyond these initial results, the integration of AI remains limited and merits further attention.

Artificial intelligence-driven engineering and evolution of gene editors

Applications of protein engineering to existing gene editors underlie some of the most considerable advances in the field, leading to the development of gene editors with enhanced functionality, specificity and expanded targeting scopes[12]. AI methods are now making this process more accessible and efficient, finding utility spanning from targeted rational design to directed evolution.

Using AI to engineer gene editors with enhanced function

AI-based protein structure prediction has enabled more effective rational engineering of gene editors in the absence of experimentally determined protein structures. The AlphaFold series of protein prediction models[18], deep learning models trained on approximately 170,000 known structures in the Protein Data Bank, have found the most use in this regard (Fig. 3a).

In a notable recent example, one study used AlphaFold2[18] to design an IscB protein with increased nucleic acid contact via insertion of nucleic acid recognition domain segments[31]. Modelling recognition domain insertions to identify those least likely to disrupt protein structure led to the identification of variants with up to 20-fold increases in editing efficiency. Further improvements came from rational mutagenesis of the top-performing variant to generate NovaIscB, which was also adapted for efficient base and epigenetic editing[31].

AlphaFold has also been valuable in generating more compact Cas variants that benefit from compatibility with adeno-associated virus (AAV) delivery systems. Using AlphaFold2 to identify regions of Cas13 unlikely to be essential for editing activity led to the development of miniaturized Cas13 variants that retained the activity of the wild-type enzyme[109]. Additionally, applying AlphaFold3[19] to model protein–RNA interactions identified and eliminated a region of a Cas12fl sgRNA detrimental to its activity, resulting in a truncated sgRNA that increased Cas12f1 cleavage activity[110].

Structural prediction methods can aid in implementing targeted alterations of genome editing enzymes that go beyond improving editing efficiency. One study identified immunogenic epitopes of different Cas proteins and used the Rosetta[111] protein prediction model, along with the deep learning model NetMHCpan 4.1 (ref. 112), to predict protein–MHC binding and redesign the epitopes for reduced immune recognition, without negatively impacting editing efficiency[113]. Another particularly notable example leveraged the RFdiffusion[114] model in conjunction with AlphaFold3[19] to design a small binding protein inhibiting the mismatch repair protein MLH1, achieving more than 2.5-fold increases in editing efficiency when used with the leading PE7 architecture[115].

Protein language models, such as evolutionary-scale models (ESMs)[71], are proving beneficial in gene editor protein engineering as well (Fig. 3b). Trained on exceptionally large protein sequence datasets, protein language models predict the likelihood of finding a given amino acid at a specific position of a protein[116]. A higher likelihood of finding an amino acid equates to an increased likelihood of that amino acid substitution benefiting protein function. In the context of genome editing, He et al. combined several ESM models to improve the activity of a uracil-*N*-glycosylase with T-to-S(G/C) functionality, subsequently generating a novel BE that expands the editing scope of base editing tools[117]. Here, ESM models were used to identify protein variants predicted to have the greatest fitness, resulting in about 2-fold increases in editing efficiency. A similar approach used multiple ESM models to identify evolutionarily likely variants within the TadA-8e deaminase, enabling the identification of variants with reduced bystander editing[118].

Lastly, prediction models trained by experimental high-throughput screening can be applied to Cas protein engineering. By screening more than 1,000 SpCas9 enzymes with varying PAM preferences, the machine learning model PAMmla was developed to predict the optimal SpCas9 amino acid substitutions for recognition of a given PAM sequence[119]. Tailored SpCas9 variants generated by the model for specific PAMs notably outperformed variants previously developed for broad PAM recognition.

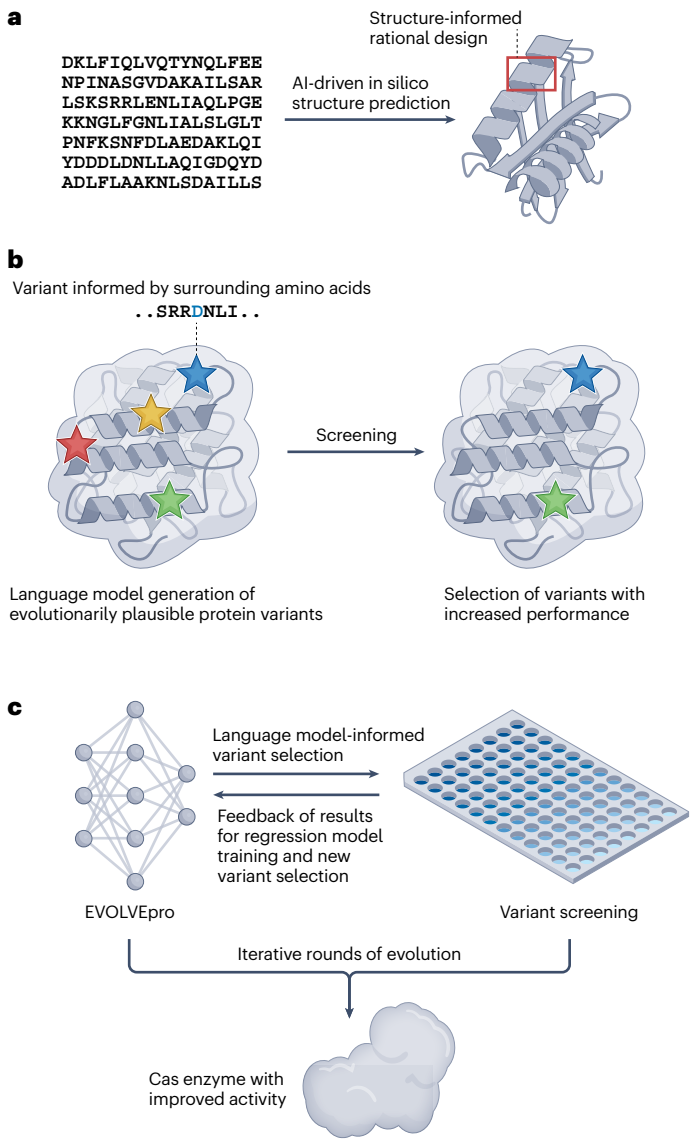

Fig. 3 | Artificial intelligence-assisted protein engineering of gene editors. **a**, Artificial intelligence (AI) methods such as AlphaFold have enabled the prediction of high-quality protein structures based on the amino acid sequence, thereby facilitating more effective rational design. **b**, Protein language models can identify amino acid substitutions that are more evolutionarily likely, suggesting variants with increased function. These can be screened to identify beneficial variant combinations. **c**, The integration of AI with directed evolution strategies, as seen with EVOLVEpro, can aid in the development of gene editors with increased editing efficiency over a small number of evolutionary rounds.

Artificial intelligence-assisted evolution of genome editing proteins

Directed evolution is an exceptionally powerful tool for gene editor engineering, with successes including the development of ABEs[37], Cas enzymes with enhanced editing efficiencies[120] and improved RTs for enhanced PEs[48]. However, directed evolution methods often require extensive experimental labour and may not capture the full landscape of possible protein variants[121]. Integrating AI with directed evolution

can aid this approach by predicting and learning optimal mutations to investigate during screening[122].

The utility of AI-assisted directed evolution for genome editing was first exemplified by EVOLVEpro[123] (Fig. 3c). EVOLVEpro uses the ESM-2 protein language model[71] in combination with a regression model to learn the impact of screened mutations on protein activity and identify optimal mutations over multiple rounds of evolution. Here, an initial set of random variants is first selected for experimental screening and embedded into the protein language model. The regression model then learns the relation between the embedding and protein activity, informing the selection of variants generated by the language model in subsequent rounds of evolution. EVOLVEpro demonstrated successes for genome editing across several modalities. After only 5 rounds of evolution, screening 12 mutants per round, a compact Cas12 nuclease was evolved to have an approximately 5-fold increase in editing activity, notably outperforming Cas12 variants previously engineered by rational design[123]. In the same study, EVOLVEpro also presented the first case of using AI to develop a more effective PE, achieving an approximately 1.5-fold increase in editing efficiency over seven rounds of evolution. This approach was further successful in improving PE-integrase systems, evolving a Bxb1 integrase with an approximately 2.6-fold increase in integration efficiency after nine rounds of evolution. Such improvements over a relatively small number of evolutionary rounds suggest considerable potential for AI-based evolution strategies in improving genome editing tools.

Artificial intelligence for the discovery and generation of novel gene editors

Substantial diversity exists across microbial and archaeal metagenomes. AI advances are now enabling more efficient searching of large biological datasets, facilitating the discovery of new enzymes with genome editing potential. Concurrently, major strides in applying language models to protein and genome design are proving useful to generate editing enzymes de novo, providing a vast source of potential genome editing tools beyond those that exist in nature.

Discovery of novel editing enzymes

Mining for novel genome editing systems among microbial datasets has typically involved sequence-based search methods based on homology with known genome editing enzymes or domains[124]. Although these methods have had great success, searching among protein families with limited sequence homology poses challenges. By searching based on similarity in structures from protein-coding genes predicted by AlphaFold, novel enzymes can be discovered that would be missed otherwise (Fig. 4a). For example, AlphaFold2 was used to search predicted structures for the presence of a pair of DNA-binding domains characteristic of Cas13 enzymes[14], uncovering an ancestral clade of Cas13 enzymes with RNA-targeting ability and a smaller size. Another approach incorporated AlphaFold2 to search for proteins containing structures resembling the Cas9 sgRNA interaction domain[125]. This approach resulted in the discovery of a new family of compact RNA-programmable nuclease systems termed TIGR-Tas (tandem-interspersed guide RNA (TIGR) associated (Tas)), which demonstrated genome editing activity in human cells. Combining structural and sequence-based search methods also promoted the characterization of Fanzor proteins as the first reported eukaryotic RNA-guided DNA endonucleases[126,127].

Structure-based search methods have proven useful also in the context of base editing. AlphaFold2 was used to predict and classify the

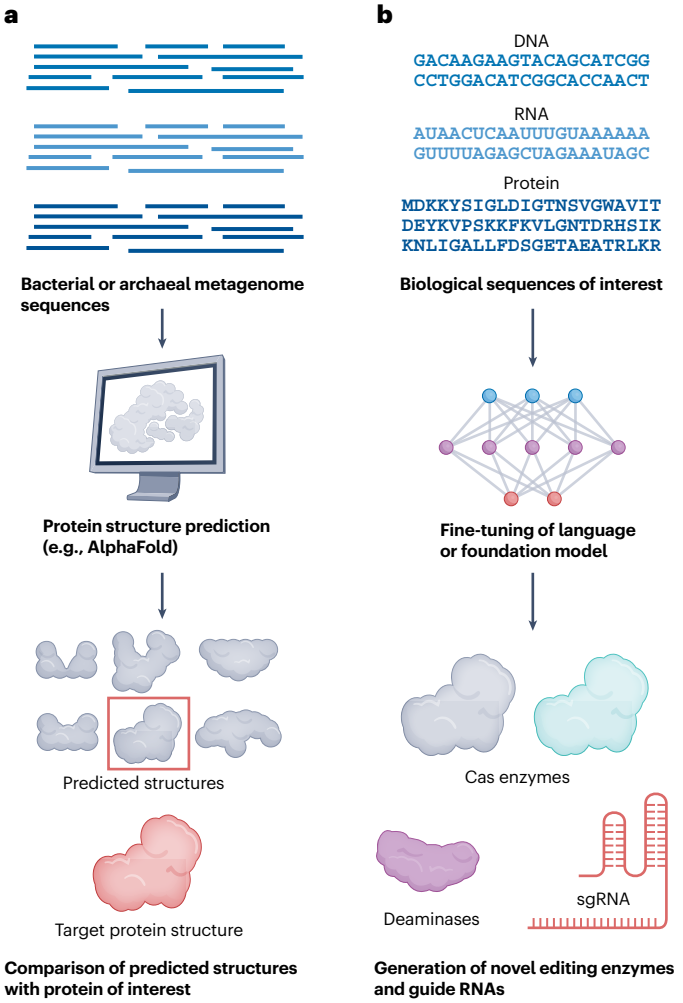

Fig. 4 | Applications of artificial intelligence for the discovery and generation of novel gene editors. a, Utilizing protein structure prediction models to search metagenomic data for proteins resembling gene editors can identify enzymes with genome editing utility that may otherwise be difficult to find via traditional nucleotide searches. **b**, Language models trained on biological data and fine-tuned on specific targets such as CRISPR operons and relevant editing enzymes can enable the de novo generation of gene editors not known to exist in nature. sgRNA, single guide RNA.

structures of 238 protein sequences from 16 families of deaminases, promoting the discovery of new smaller-sized cytidine deaminases[128]. These deaminases exhibited unique sequence preferences and exhibited higher on-target editing and reduced off-target editing relative to a standard rAPOBEC1 deaminase. Similarly, AlphaFold2 was used to predict the structures of proteins containing deaminase domains, using 1,438 deaminases from the APOBEC-like family[129]. This approach facilitated the discovery of cytidine deaminases with high editing efficiencies comparable with high-efficiency TadA-derived standards, several of which notably exhibited no sequence preference, showing equivalent activity at AC, TC, CC and GC motifs.

For CRISPR protein families with few known homologues, protein language models offer a promising approach to identify novel Cas enzymes from microbial sequence datasets. The CHOOSER framework

Table 1 | Artificial intelligence models for the optimization, engineering and discovery of genome editing modalities

AI model	Task	AI method	Refs.
Nuclease sgRNA design			
DeepCRISPR	Cas9 sgRNA on-target and off-target prediction	Deep learning, CNN	67
DeepSpCas9	Cas9 sgRNA on-target prediction	Deep learning, CNN	66
DeepCpf1	Cas12a sgRNA on-target prediction	Deep learning, CNN	65
CRISPRon	Cas9 sgRNA on-target prediction	Deep learning, CNN	68
CRISPR_HNN	Cas9 sgRNA on-target prediction	Deep learning, CNN, bidirectional gated recurrent unit	69
PLM-CRISPR	Cas9 variant sgRNA on-target prediction	Deep learning, CNN, protein language model (ESM-2)	70
DeepCRISTL	Cas9 sgRNA on-target prediction	Deep learning, transfer learning	72
TnpB editing efficiency predictor (TEEP)	TnpB ωRNA on-target prediction	Deep learning, CNN, RNN	29
DeepCas13	Cas13d sgRNA on-target prediction	Deep learning, CNN, RNN	78
TIGER	Cas13d sgRNA on-target prediction	Deep learning, CNN	79
CnnCRISPR	Cas9 sgRNA off-target prediction	Deep learning, CNN	73
CRISOT	Cas9 sgRNA off-target prediction	Machine learning, XGBoost	74
crispAI	Cas9 sgRNA off-target prediction with uncertainty estimates	Deep learning, CNN	75
CRISPR-Embedding	Cas9 sgRNA off-target prediction	Deep learning, CNN, *k*-mer clustering	76
CCLMoff	Cas9 sgRNA off-target prediction	Deep learning, CNN, RNA language model	77
Prediction of DNA repair outcomes following Cas9 cleavage			
inDelphi	DNA sequence outcome prediction following DSB repair	Machine learning, linear regression, neural networks	80
Lindel	DNA sequence outcome prediction following DSB repair	Machine learning, logistic regression	81
FORECasT	DNA sequence outcome prediction following DSB repair	Machine learning, logistic regression	82
SPROUT	DNA sequence outcome prediction following DSB repair	Machine learning, gradient boosting	83
Base editing sgRNA design			
BE-Hive	Prediction of base editing outcomes	Efficiency prediction: machine learning, gradient-boosted regression trees Bystander editing: deep learning, deep conditional autoregressive model	86
DeepABE/DeepCBE	Prediction of base editing outcomes	Deep learning, CNN	87
BE-Dict	Prediction of base editing outcomes	Deep learning, attention-based transformer architecture	88
CGBE-Hive	Prediction of base editing outcomes	Efficiency prediction: machine learning, gradient-boosted regression trees Bystander editing: deep learning, deep conditional autoregressive model	89
DeepBE, DeepCas9variants	Prediction of base editing outcomes, identification of optimal Cas9 variants for different edit sites	Deep learning, CNN	90
BEguider	Prediction of base editing outcomes for near-PAM-less BEs	Deep learning, CNN	91
ABEdeepoff, CBEdeepoff	BE off-target prediction	Deep learning, RNN	94
Prime editing pegRNA design			
DeepPE	pegRNA optimization for +5 G-to-C edit	Deep learning, CNN	95
EasyPrime	pegRNA optimization for single base changes	Machine learning, XGBoost	96
PRIDICT	pegRNA optimization for single base changes, small indels	Deep learning, RNN	97
DeepPrime	pegRNA optimization for single base changes, small indels	Deep learning, CNN	98
PRIDICT2.0	pegRNA optimization with improved performance for edits up to 15 bp	Deep learning, RNN	99
ePRIDICT	Influence of chromatin context on editing efficiency	Machine learning, XGBoost	99
MinsePIE	pegRNA insertion efficiency up to 69 bp	Machine learning, XGBoost	100
Optimized Prime Editing Design (OPED)	pegRNA optimization for single base changes, insertions and deletions	Nucleotide language model	101
DTMP-Prime	pegRNA optimization for single base changes, insertions, deletions and off-target prediction	Deep learning, transformer model	102

Table 1 (continued) | Artificial intelligence models for the optimization, engineering and discovery of genome editing modalities

AI model	Task	AI method	Refs.
Epigenome editing sgRNA activity prediction			
EpiCas-DL	sgRNA prediction for CRISPRa, CRISPRi and CRISPRoff	Deep learning, CNN	105
Launch-dCas9	Prediction of CRISPRi sgRNA outcomes	Deep learning, CNN; machine learning, XGBoost	106
Gene editor protein engineering			
AlphaFold2	Protein structure prediction	Deep learning, transformer architecture	18
AlphaFold3	Protein structure prediction, diverse protein–biomolecule interactions (ligands, nucleic acids, covalent modifications)	Deep learning, generative AI, diffusion model	19
Rosetta with NetMHCpan 4.1	Epitope binding affinity to protein structures	NetMHCpan 4.1: deep learning	111,112
RFdiffusion	Protein structure design	Deep learning, generative AI, diffusion model	114
ESM-2	Identifying evolutionary-likely protein variants	LLM	71
PAMmla	Custom PAM variant design	Machine learning, neural network	119
EVOLVEpro	Directed evolution	LLM (ESM-2) and random forest regression model	123
Gene editor discovery and de novo generation			
EVO	De novo generation of DNA, RNA and proteins	Foundation model, LLM	131
ProGen2	De novo generation of proteins	LLM	132
AlphaFold2	Protein structure prediction	Deep learning, transformer architecture	18
CHOOSER	CRISPR–Cas system discovery	LLM (ESM-2) trained on Cas protein sequences	130
FLSHclust (fast locality-sensitive hashing)	Sequence-based protein clustering	Deep learning-assisted clustering	15
CasGen	De novo generation of Cas proteins	Deep learning, transformer architecture, generative AI	134
Automated experimental design			
CRISPR-GPT	Design of CRISPR genome editing experiments	LLM	145
Biomni	Experimental design and guidance for general biomedical research, including genome editing	LLM	146

AI, artificial intelligence; BE, base editor; CNN, convolutional neural network; CRISPRa, CRISPR activation; CRISPRi, CRISPR inhibition; DSB, double-strand break; indels, insertions or deletions; LLM, large language model; PAM, protospacer adjacent motif; pegRNA, prime editing guide RNA; RNN recurrent neural network; sgRNA, single guide RNA.

exemplifies this strategy: by fine-tuning the ESM-2 protein language model on Cas9, Cas12 and Cas13 protein sequences[130], CHOOSER facilitated the discovery of 39 new Cas12 homologues, including 11 Casλ homologues of phage origin. Biochemical characterization of one Casλ homologue confirmed its double-stranded DNA cleavage activity.

AI can further enhance the discovery of new genome editing systems by enabling the development of more effective search algorithms to analyse the increasingly large microbial datasets being generated. The deep learning-assisted clustering algorithm FLSHclust (fast locality-sensitive hashing) is capable of rapidly clustering protein sequences and working with database sizes of up to billions of proteins[15]. Applying FLSHclust to a curated database containing 8.8 Tbp of prokaryotic genomes (8 billion proteins and 10.2 million CRISPR arrays) led to the discovery of 188 new CRISPR-associated systems, 4 of which were chosen for biochemical characterization and shown to have DNA or RNA cleavage activity.

De novo generation of gene editors

The generation of entirely new gene editors presents a novel direction for gene editor 'discovery'. Major strides in applying language models to protein and genome design are proving useful to generate editing enzymes de novo (Fig. 4b). A pioneering example of applying language models to genome editing comes from the genomic foundation model EVO[131], which is trained on 2.7 million prokaryotic and phage genomes, and is capable of zero-shot prediction and generation tasks across DNA, RNA and protein modalities. When EVO was refined on 72,831 CRISPR operon sequences, it generated a novel Cas9 protein (EvoCas9-1) and sgRNA with editing efficiency comparable with the standard SpCas9. Interestingly, the generated EvoCas9-1 sgRNA also improved the in vitro cleavage activity of SpCas9. Another notable application fine-tuned the ProGen2 protein language model[132] on approximately 1.25 million CRISPR operon sequences[133]. The refined model generated novel sequences for CRISPR–Cas enzymes with a reported 4.8-fold increase in sequence diversity relative to the natural sequences and AlphaFold-predicted protein structures consistent with their respective enzyme families. Further fine-tuning on 238,917 Cas9 sequences resulted in the generation of a Cas9 enzyme (OpenCRISPR-1) with editing efficiency comparable with SpCas9, along with the benefit of substantially reduced off-target editing.

Language models have been used to help generate novel base editing enzymes as well. In the OpenCRISPR study, the authors concurrently generated novel adenine deaminases by training the ProGen2

model on a dataset of TadA deaminase enzymes[133]. ABEs using these novel deaminases exhibited editing efficiencies comparable with the highly effective ABE8.20. The OpenCRISPR-1 Cas9 protein could also function as a BE when fused with a deaminase, highlighting how AI-generated Cas variants can be readily adapted to more advanced genome editing systems.

Box 2 | Artificial intelligence-powered virtual cells for in silico prediction of CRISPR perturbation outcomes

CRISPR screening has transformed functional genomics by enabling systematic, targeted perturbation of genes across diverse cell types and organisms[177]. Incorporating single-cell profiling technologies[178] into CRISPR screening workflows has further expanded their utility, enabling causal analysis of gene function and cellular states by linking individual genetic perturbations to transcriptomic consequences at single-cell resolution. Notable examples include Perturb-Seq[179], CRISP–seq[180] and CROP-seq[181]. Together with existing observational cell atlases, these large-scale perturbational datasets are driving the development of virtual cell models, which are deep learning frameworks capable of capturing molecular interactions, simulating cell functions and predicting transcriptomic, phenotypic or morphological changes induced by natural development or external interventions[141,142,144].

Virtual cell perturbation models for predicting CRISPR outcomes

One of the most promising applications of virtual cell models is the in silico prediction of transcriptomic responses to genetic perturbations[142–144]. By training on large-scale genetic interventional datasets such as Perturb-seq, these models can generalize to predict cellular responses for unseen genes, cell types or perturbation modalities. This capability enables researchers to computationally screen numerous interventions, generate testable hypotheses and, thereby, accelerate the identification of underlying cellular mechanisms and new therapeutics.

Existing efforts in this space began with bespoke models that preceded the rise of virtual cell frameworks. For example, GEARS (graph-enhanced gene activation and repression simulator) is a graph-based model that integrates gene interaction networks to predict responses to unseen single-gene and multigene perturbations[182]. PerturbNet is a deep generative model that predicts distributions of single-cell expression profiles[183]. Additionally, CREME (*cis*-regulatory element model explanations) functions as a virtual laboratory, using deep neural networks trained on gene regulation data to predict how perturbing genomic regions influences gene expression[184]. By contrast, GPerturb, a simpler Bayesian model, emphasizes interpretability and provides uncertainty estimates while maintaining competitive predictive performance compared with deep learning models[185].

A promising strategy to overcome the challenges of building accurate virtual cell models is the development of multimodal foundation models[144]. These models are trained in a self-supervised manner on large, multimodal datasets, enabling them to learn foundational representations of cell states and gene regulatory dynamics. Once fine-tuned on perturbation datasets, they can generalize across unseen genes, cell types or perturbation modalities. For example, scGPT is a transformer-based model trained on more than 33 million single-cell profiles and fine-tuned for transcriptomic response prediction tasks[186]. Beyond modelling known perturbation outcomes, scGPT can also infer underlying perturbations that give rise to specific cell states, an ability that supports the identification of potential CRISPR perturbation targets for disease modulation.

Another prominent example is Geneformer, a transformer-based model pretrained on approximately 30 million single-cell transcriptomes across diverse tissues[187]. When fine-tuned on perturbation data, Geneformer demonstrated strong generalization ability, even in low-data settings. Notably, it successfully predicted that TEAD4 knockout would impair cardiomyocyte function, a finding subsequently validated using CRISPR experiments.

The STATE model offers a further advance, which consists of two integrated submodules: a state transition model and a state embedding model[188]. Trained on 167 million cells of observational data and more than 100 million perturbational single-cell profiles, STATE accurately predicts transcriptomic responses to CRISPR knockdowns in unseen cell lines. It outperformed linear baselines across key metrics, including gene expression counts, differential gene expression and the perturbation effect size.

Several other foundation models contribute to this growing ecosystem. scFoundation, trained on 50 million single-cell profiles, produces general-purpose embeddings that support downstream perturbation modelling[189]. CellFM, a large-scale foundation model pretrained on single-cell transcriptomics of 100 million human cells using a modified RetNet framework, has demonstrated superior performance in predicting perturbation responses[190].

Spatially resolved CRISPR perturbation modelling

An emerging frontier in virtual perturbation modelling is the integration of spatial context. Recent advances in spatial CRISPR screening techniques, such as Perturb-map[191], Perturb-DBiT[192], Perturb-Multi[193] and Perturb-FISH[194], combine CRISPR perturbations with spatial transcriptomics to map the effects of specific single guide RNAs (sgRNAs) directly within native tissue environments. These methods preserve spatial structure, allowing researchers to assess how gene perturbations influence cell-intrinsic gene expression changes, but also the perturbation-induced alterations in tissue organization and intercellular communication.

Although artificial intelligence (AI) application in this domain remains nascent, promising developments are beginning to surface. One example is Celcomen, a generative graph neural network designed to disentangle intracellular and intercellular gene regulatory programmes in spatial and single-cell transcriptomic data[195]. By learning both gene–gene interactions and cell–cell communication dynamics, Celcomen can simulate spatial transcriptomic responses to genetic perturbations. More broadly, emerging AI tools for spatial transcriptomics are likely to be adapted for CRISPR-focused modelling[196]. As experimental and computational methods in this area continue to advance, spatially informed virtual perturbation models are poised to become powerful tools for decoding gene function in complex biological systems.

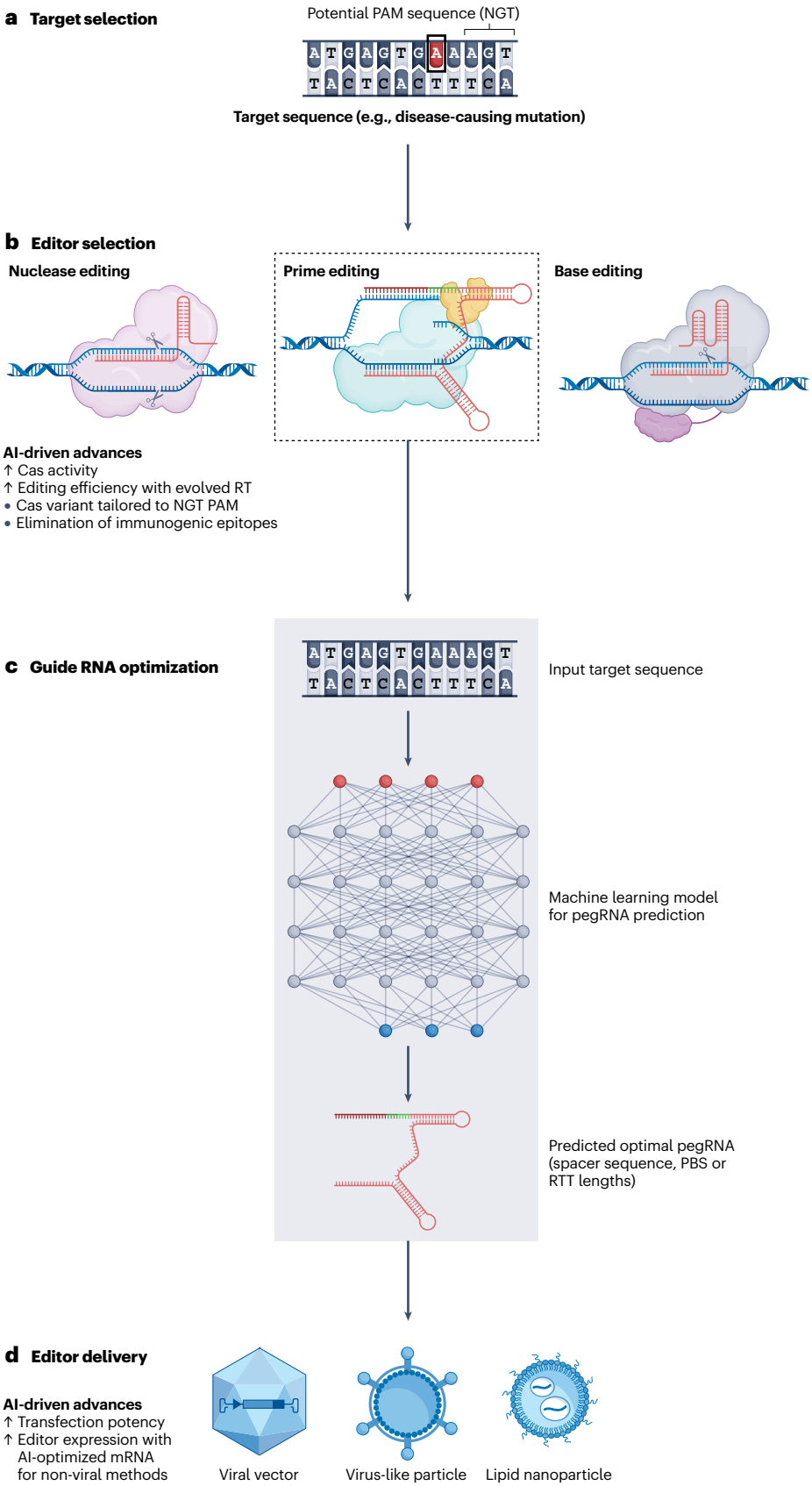

Fig. 5 | How artificial intelligence methods can be integrated into the genome editing pipeline for a specific disease target. a, In this case, we start with the aim of correcting a disease-causing mutation indicated by 'A' (red). **b**, With the target selected, an appropriate editor for gene correction needs to be chosen. As we are looking to precisely change a single base, nuclease editing is not applicable. Furthermore, the presence of multiple 'A' mutations nearby brings the risk of bystander editing with base editing. The prime editor (PE) is therefore chosen, which can be further improved by the noted artificial intelligence (AI)-driven enhancements and tailored specifically to the nearby NGT protospacer adjacent motif (PAM). **c**, With an editor chosen, the prime editing guide RNA (pegRNA) will need to be optimized for the target, a process that can be made more efficient using machine learning prediction models. **d**, Finally, if in vivo delivery is desired, viral and non-viral delivery vehicles can be utilized. Here, AI can aid in the generation of delivery vehicles with enhanced transfection potencies and mRNA with increased editor expression. PBS, primer binding site; RT, reverse transcriptase; RTT, RT template.

Glossary

Bidirectional gated recurrent unit
A type of recurrent neural network (RNN) model that learns patterns in sequences, reading them in both forward and backward directions, to help capture context from past and future information.

Convolutional neural networks
(CNNs). Deep learning models that process data with a grid-like structure, such as images or sequences, by applying convolutional filters to learn spatial patterns and hierarchical feature representations.

Deep learning
A branch of machine learning that uses multilayered artificial neural networks to automatically learn latent representations of data for tasks such as prediction, classification and generation.

Foundation model
A large-scale pretrained model built on diverse data that can be adapted to a wide range of downstream tasks with minimal task-specific tuning.

***k*-mer embeddings**
Vector representations of fixed-length sub-sequences (*k*-mers) from biological sequences that capture their contextual and semantic properties for use in machine learning models.

Language model
A deep learning model that learns the statistical patterns of a language or biological sequence to predict and generate new text or sequences.

Logistic regression
A machine learning method used to model the probability of a specific outcome, such as single guide RNA (sgRNA) activity, based on user-supplied features.

Multi-head self-attention
A mechanism that projects input sequences into multiple attention heads, each learning different relationships among tokens, and then combines them to capture diverse contextual information.

Multiscale convolution
A neural network technique that applies convolutional filters of different kernel sizes to the same input so that the model can capture both local and global features simultaneously.

One-hot encoding
A representation method that converts categorical variables into binary vectors where only the position corresponding to the category is set to 1 and all others are 0.

sgRNA activity
The efficiency with which a given single guide RNA (sgRNA) directs its associated nuclease to induce edits. Typically measured as the frequency of insertions or deletions (indels) at the target site.

SpRY Cas9 variant
Developed through protein engineering of the protospacer adjacent motif (PAM)-interacting domain of SpCas9, this variant recognizes NRN or NYN PAMs (where R = A/G, Y = C/T), effectively allowing nearly any sequence in the genome to be targeted.

Support vector machine
A machine learning technique used to classify data into different categories (for example, high versus low-performing single guide RNA (sgRNA)) based on user-provided features.

Transformer
A deep learning model based on self-attention mechanisms that process input sequences while efficiently capturing long-range dependencies.

Zero-shot prediction
The ability of a model to make predictions on tasks or classes it was never explicitly trained on by leveraging generalized knowledge learned during pretraining.

Beyond language model-based approaches, CasGen[134] uses a transformer architecture[135] trained on 3,021 Cas9 and 287 Cas12 protein sequences to generate novel Cas variants. A BLAST-based sequence alignment filter is applied to exclude outputs with high similarity to known enzymes. AlphaFold predictions of the generated sequences showed high structural similarity to existing Cas proteins[132], although experimental validation remains pending.

Conclusions and perspectives

AI is rapidly transforming the field of genome editing by accelerating target optimization, enabling the discovery of novel enzymes and improving the engineering of existing modalities (Table 1).

One immediate direction for future research is the application of AI beyond the CRISPR-based platforms discussed in this Review. For instance, transposon-based systems[136–138] and recently discovered bridge RNAs[139,140] are emerging as promising tools for site-specific insertion of large genetic payloads. Optimizing these systems will require deciphering how the sequence context influences integration efficiency, an inherently complex and high-dimensional problem well suited to AI-driven modelling. Furthermore, given the success of AI in protein discovery and engineering, expanding such strategies to these novel genome editing platforms should yield more advances.

Although this Review primarily focuses on AI-driven approaches for engineering and optimizing CRISPR-based genome editors, the computational prediction of cellular responses to CRISPR perturbations is a rapidly emerging synthesis of AI and CRISPR technologies poised to have a substantial impact on biomedical research[141] (Box 2). Training AI models on vast single-cell CRISPR perturbation datasets is enabling researchers to construct virtual cell models to predict downstream transcriptomic and phenotypic outcomes in response to user-specified perturbations[142–144]. The ability to virtually screen CRISPR perturbations across diverse cell and tissue types has the potential to greatly accelerate the discovery of novel cellular mechanisms and therapeutic strategies.

Recent advances in large language models have also enabled AI agents to automate multiple stages of genome editing experimental design. A pioneering example is CRISPR-GPT, which integrates information from open-access discussions, experimental protocols and published literature to assist users throughout the entire genome editing workflow, from Cas selection and sgRNA design to data analysis[145]. Notably, the authors reported that CRISPR-GPT enabled two junior researchers with no prior gene editing experience to successfully carry out multigene knockouts and CRISPRa experiments. At a broader scale, Biomni is a general-purpose AI agent for biomedical research that integrates more than 150 tools to support hypothesis generation,

experimental protocol design and bioinformatic analyses[146]. Within the context of genome editing, Biomni can automate essential tasks including sgRNA library design, Cas9 editing simulations, mutational outcome analysis and the design of CRISPR perturbation screens, allowing researchers to execute entire workflows through natural language prompts. These early platforms highlight the potential of AI agents to enhance experimental efficiency and democratize access to genome editing technologies. Looking ahead, integrating AI agents with self-driving laboratories[147] could enable fully autonomous systems that design, execute and analyse genome editing experiments, requiring only minimal input from researchers to define the experimental goals.

From a translational perspective, therapeutic genome editing represents one of the field's most prominent goals, with several CRISPR-based clinical trials currently underway[23,39]. The recent approval of a CRISPR-based therapy for sickle cell disease marked a historic milestone[148]. However, current ex vivo strategies address only a narrow range of indications. Broadening the clinical utility of genome editing will require the development of safe and efficient in vivo delivery systems capable of targeting diverse tissues[149]. AI is already showing promise in aiding the design of delivery vehicles, including viral vectors[150], virus-like particles[151] and non-viral platforms such as lipid nanoparticles[152–157]. For example, the LUMI model[152], a foundation model pretrained on more than 28 million molecules, was recently developed to guide the discovery of novel lipid nanoparticles for in vivo delivery of an RNA-based Cas9 gene editor. In parallel, AI-assisted mRNA optimization[158–161] is a rapidly growing field that remains under-used in genome editing. Incorporating AI-based mRNA sequence and structure design into genome editing workflows could synergize with delivery innovations, further enhancing editing specificity and efficiency. We envision that incorporating all these advances into genome editing workflows will enable a future standardized pipeline readily adaptable to different disease-causing mutations (Fig. 5). Furthermore, AI methods can support clinical translation by facilitating the identification of pathogenic variants[162,163], linking patient phenotypes to causal genotypes[164,165] and informing clinical trial design and implementation[166].

To fully realize the promise of AI in genome editing, improvements in data quality and infrastructure will be essential. Despite the proliferation of AI models, their success remains limited by the availability of large and high-quality datasets. In areas such as sgRNA design, most data are generated by individual laboratories using varied experimental protocols and annotation schemes. This heterogeneity hampers model generalization and training. Establishing shared standards and collaborative databases, similar to the Protein Data Bank[167,168] and CASP (critical assessment of protein structure prediction) competitions that enabled AlphaFold, could catalyse progress across genome editing applications.

Beyond data quality, many of the discussed topics relate to searches in a vast space of possibilities. For example, the number of potential protein sequences of 300 amino acids exceeds 20^{300}, a scale that cannot be exhaustively explored even with large datasets. Improving data efficiency is therefore paramount. Self-driving laboratories and active learning strategies offer a powerful solution to the challenge of navigating vast design spaces[147,152,169,170]. Rather than relying on static model training, these approaches iteratively guide experimentation by selecting the most informative data points for validation.

In summary, AI is readily transforming the landscape of genome editing by enabling systematic navigation of complex design spaces, enhancing experimental efficiency and supporting translational breakthroughs. As the field continues to progress, the continued integration of AI into its development will mark a pivotal step towards achieving precision, scalability and therapeutic impact.

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